

NAME: MUSKAN SHAH

PRN: 23070122279

BATCH:2023-2027 (Devops L4)

LAB ASSINGMENT: 1.3

Basic Containerization (Docker) & Jenkins Freestyle Project)

```
Jul 24 2:52 PM
muskan123@muskan:~/Documents
~/Documents

muskan123@muskan ~/Documents
> sudo dnf install docker
[sudo] password for muskan123:
Updating and loading repositories:
Repositories loaded.
Failed to resolve the transaction:
Problem: problem with installed package
- installed package docker-ce-cli-1:29.6.2-1.fc44.x86_64 conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.3.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.4.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.4.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.4.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.4.2-2.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.4.3-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.5.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.5.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.5.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.5.3-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.6.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.6.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- package docker-ce-cli-1:29.6.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.3.0-1.fc44.x86_64 from fedora
- conflicting requests
- installed package docker-ce-cli-1:29.6.2-1.fc44.x86_64 conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.3.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.4.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.4.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.4.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.4.2-2.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.4.3-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.5.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.5.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.5.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.5.3-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.6.0-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.6.1-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
- package docker-ce-cli-1:29.6.2-1.fc44.x86_64 from docker-ce-stable conflicts with docker provided by moby-engine-29.6.0-1.fc44.x86_64 from updates
You can try to add to command line:
--allowrasing to allow removing of installed packages to resolve problems
--skip-broken to skip uninstallable packages
muskan123@muskan ~/Documents
> sudo dnf install jenkins
Updating and loading repositories:
Repositories loaded.
Failed to resolve the transaction:
No match for argument: jenkins
You can try to add to command line:
--skip-unavailable to skip unavailable packages
muskan123@muskan ~/Documents
>
```

```
Jul 24 3:31 PM
muskan123@muskan: ~/Documents — docker pull jenkins/jenkins:lts-jdk17
~/Documents

Sorry, try again.
[sudo] password for muskan123:
No packages to remove for argument: moby-engine
No packages to remove for argument: moby-filessystem

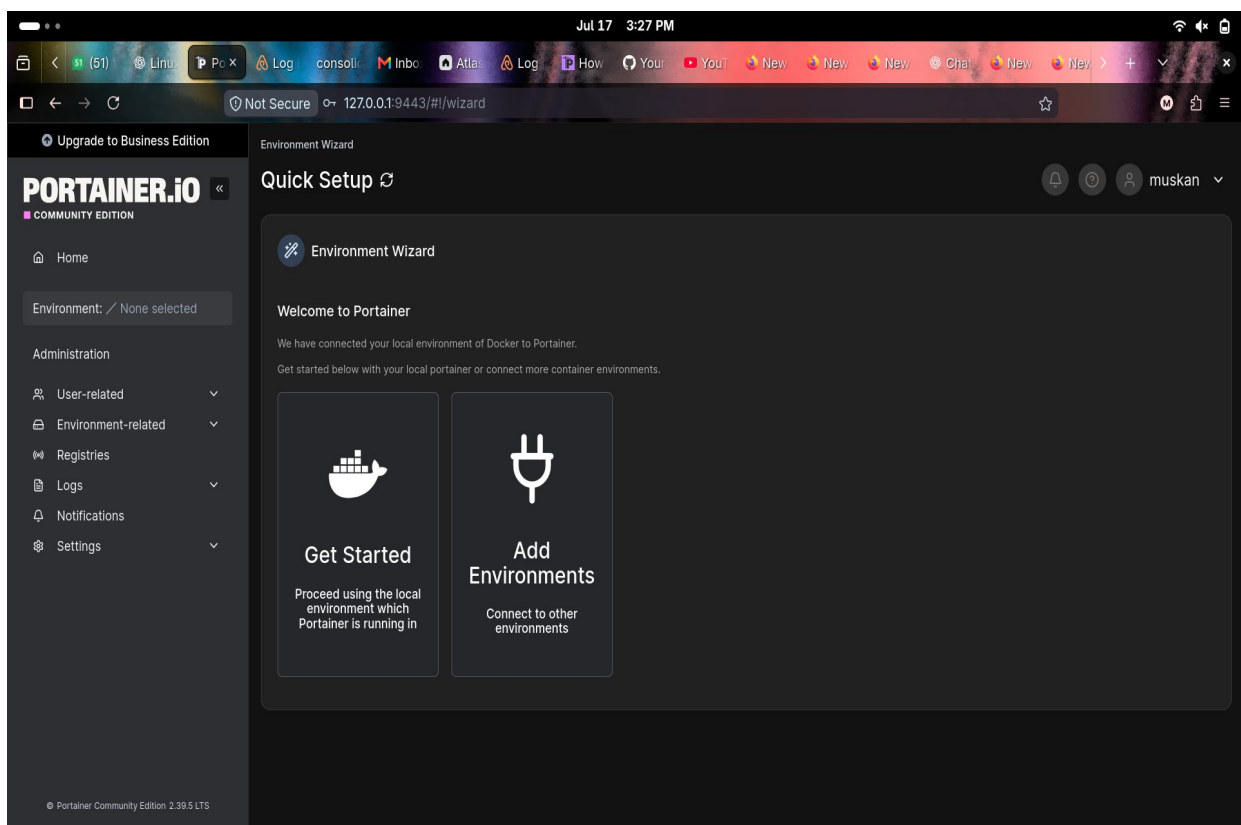
Nothing to do.
muskan123@muskan ~/Documents
> sudo systemctl enable docker
muskan123@muskan ~/Documents
> sudo systemctl start docker
muskan123@muskan ~/Documents
> docker --version
Docker version 29.6.2, build dfc4efb
muskan123@muskan ~/Documents
>
> muskan123@muskan ~/Documents
> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
7fc3e17ba64f   portainer/portainer-ce:latest      "/portainer"            7 days ago    Up 37 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443
muskan123@muskan ~/Documents
> docker pull jenkins/jenkins:lts-jdk17
lts-jdk17: Pulling from jenkins/jenkins
53c88f1dfcb7: Downloading [=====] 15.73MB/49.29MB
4d2125ccf71c: Downloading [=====] 16.78MB/59.45MB
69583609dc01: Download complete
0b9e958129f4: Download complete
46f93bf1cb2f: Download complete
01938f7c040b: Download complete
503dee4523cf: Downloading [=====] 15.73MB/63.45MB
76eac8d3a597: Download complete
6da7907f78f4: Download complete
6327a7366a6f: Download complete
f44eb945cec1: Downloading [=====] 17.83MB/96.12MB
b7b154fd69a1: Download complete
c47cc6af1e21: Download complete
```

```
Jul 24 3:39 PM
muskan123@muskan: ~/Documents
~/Documents

> docker pull jenkins/jenkins:lts-jdk17
lts-jdk17: Pulling from jenkins/jenkins
53c88f1dfcb7: Pull complete
4d2125ccf71c: Pull complete
69583609dc01: Pull complete
0b9e958129f4: Pull complete
46f93bf1cb2f: Pull complete
01938f7c040b: Pull complete
503dee4523cf: Pull complete
76eac8d3a597: Pull complete
6da7907f78f4: Pull complete
6327a7366a6f: Pull complete
f44eb945cec1: Pull complete
b7b154fd69a1: Pull complete
c47cc6af1e21: Download complete
Digest: sha256:ccc1eab79902722f3cf7a8831c41e95c39f4a42b9a2981959630cc66539b3c2d
Status: Downloaded newer image for jenkins/jenkins:lts-jdk17
docker.io/jenkins/jenkins:lts-jdk17
muskan123@muskan ~/Documents
> docker run -d -p 8080:8080 -p 50000:50000 -v jenkins_home:/var/jenkins_home --name myjenkins jenkins/jenkins:lts-jdk17
ac279514177faa29986c55587e938d0089238f8821b367b0047dd8e48077b062
muskan123@muskan ~/Documents
> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
ac279514177f   jenkins/jenkins:lts-jdk17          "/usr/bin/tini -- /u-  9 seconds ago  Up 8 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:50000->50000
/tcp, [::]:50000->50000/tcp
myjenkins
7fc3e17ba64f   portainer/portainer-ce:latest      "/portainer"            7 days ago    Up 48 minutes  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp, 0.0.0.0:9443->9443/t
cp, [::]:9443->9443/tcp, 9000/tcp
portainer
muskan123@muskan ~/Documents
> docker exec myjenkins cat /var/jenkins_home/secrets/initialAdminPassword
cat: /var/jenkins_home/secrets/initialAdminPassword: No such file or directory
muskan123@muskan ~/Documents
> Install Suggested Plugins
bash: Install: command not found...
Similar command is: 'install'
muskan123@muskan ~/Documents
>
```

```
Jul 17 3:26 PM
muskan123@muskan:~
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:358 > encryption key file not present | filename=/run/secrets/portainer
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:398 > proceeding without encryption key |
2026/07/17 09:52AM INF github.com/portainer/portainer/api/database/boltdb/db.go:137 > loading PortainerDB | filename=portainer.db
2026/07/17 09:52AM INF github.com/portainer/portainer/api/internal/ssl/ssl.go:79 > no cert files found, generating self signed SSL certificates |
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:256 > no administrator account configured; admin initialization and backup restore require this setup token in the X-Setup-Token header. Start with --no-setup-token to disable. | setup_token=74c4e824b0f41fbdec8cb39a6b12da25c6d707310b61f5e1b9391a80ec1fbdc6
2026/07/17 09:52AM INF github.com/portainer/portainer/api/chisel/service.go:228 > generated a new Chisel private key file | private-key=/data/chisel/private-key.pem
2026/07/17 09:52:37 server: Reverse tunnelling enabled
2026/07/17 09:52:37 server: Fingerprint EftHBg42Rifl2U5Kl08A19CSzflWnmL9p+grQYh/Lb8=
2026/07/17 09:52:37 server: Listening on http://0.0.0.0:8000
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:678 > starting Portainer | build_number=29 go_version=go1.25.11 image_tag=2.39.5-linux-amd64 nodejs_version=v22.23.1 pnpm_version=10.27.0 version=2.39.5 webpack_version=5.105.0
2026/07/17 09:52AM INF github.com/portainer/portainer/api/http/server.go:371 > starting HTTPS server | bind_address=:9443
2026/07/17 09:52AM INF github.com/portainer/portainer/api/http/server.go:355 > starting HTTP server | bind_address=:9000
muskan123@muskan ~
> docker ps
CONTAINER ID        IMAGE               NAMES             COMMAND             CREATED            STATUS             PORTS
7fc3e17ba64f      portainer/portainer-ce:latest   "/portainer"      4 minutes ago      Up 4 minutes      0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9443->9443/tcp, [::]:9443->9443/tcp
muskan123@muskan ~
> docker logs portainer --tail 50
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:358 > encryption key file not present | filename=/run/secrets/portainer
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:398 > proceeding without encryption key |
2026/07/17 09:52AM INF github.com/portainer/portainer/api/database/boltdb/db.go:137 > loading PortainerDB | filename=portainer.db
2026/07/17 09:52AM INF github.com/portainer/portainer/api/internal/ssl/ssl.go:79 > no cert files found, generating self signed SSL certificates |
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:256 > no administrator account configured; admin initialization and backup restore require this setup token in the X-Setup-Token header. Start with --no-setup-token to disable. | setup_token=74c4e824b0f41fbdec8cb39a6b12da25c6d707310b61f5e1b9391a80ec1fbdc6
2026/07/17 09:52AM INF github.com/portainer/portainer/api/chisel/service.go:228 > generated a new Chisel private key file | private-key=/data/chisel/private-key.pem
2026/07/17 09:52:37 server: Reverse tunnelling enabled
2026/07/17 09:52:37 server: Fingerprint EftHBg42Rifl2U5Kl08A19CSzflWnmL9p+grQYh/Lb8=
2026/07/17 09:52:37 server: Listening on http://0.0.0.0:8000
2026/07/17 09:52AM INF github.com/portainer/portainer/api/cmd/portainer/main.go:678 > starting Portainer | build_number=29 go_version=go1.25.11 image_tag=2.39.5-linux-amd64 nodejs_version=v22.23.1 pnpm_version=10.27.0 version=2.39.5 webpack_version=5.105.0
2026/07/17 09:52AM INF github.com/portainer/portainer/api/http/server.go:371 > starting HTTPS server | bind_address=:9443
2026/07/17 09:52AM INF github.com/portainer/portainer/api/http/server.go:355 > starting HTTP server | bind_address=:9000
muskan123@muskan ~
>
```

\



Upgrade to Business Edition

PORTAINER.IO

COMMUNITY EDITION

Home

local

Dashboard

Templates

Stacks

Containers

Images

Networks

Volumes

Events

Host

Administration

User-related

Environment-related

Registries

© Portainer Community Edition 2.39.5 LTS

Containers

Container list

Containers

Search...

Start

Stop

Kill

Restart

Pause

Resume

Remove

+ Add container

Name	State	Quick Actions	Stack	Image	Created	IP Address	Published Ports	Ownership
brave_haibt	exited - code 0		-	hello-world	2026-07-17 15:02:53	-	-	administrators
fervent_chatterjee	exited - code 0		-	hello-world	2026-07-17 15:03:20	-	-	administrators
nginx	running		-	nginx:latest	2026-07-17 15:30:36	172.17.0.3	-	administrators
portainer	running		-	portainer/portainer-ce:latest	2026-07-17 15:22:36	172.17.0.2	9443:9443 8000:8000	administrators

Items per page 10

Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

Welcome to Jenkins!

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job

Set up a distributed build

Set up an agent

Configure a cloud

Learn more about distributed builds

REST API

Jenkins 2.541.3

Jul 24 3:44 PM

Lecture 5

(53) WhatsA

Sign in - Jen

Sent Mail - m

DjVu Docume

Docker Jen

New Item x

Your Repo

Jira

404 Not Foun

Microsoft V

+

✓

✕

←

→

↺

http://localhost:8080/view/all/newJob

☆

Jenkins

All

New Item

🔍

⚙️

👤

New Item

Enter an item name

my-CICD-pipeline

Select an item type

**Freestyle project**
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

**Folder**
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.

**Organization Folder**
Creates a set of multibranch project subfolders by scanning for repositories.

OK

REST API

Jenkins 2.541.3

Jul 24 4:13 PM

Lecture 5

(53) WhatsA

Sign in - Jen

Sent Mail - m

Microsoft W

Docker Jen

My-CICD- x

saahmusker

Jira

New Tab

+

✓

✕

←

→

↺

http://localhost:8080/job/My-CICD-Pipeline/2/console

80%

☆

Jenkins

My-CICD-Pipeline

#2

🔍

⚙️

👤

```
[Pipeline] stage
[Pipeline] { (Build)
[Pipeline] sh
+ echo Building Project...
Building Project...
[Pipeline] sh
+ ls -la
total 16
drwxr-xr-x. 1 jenkins jenkins 26 Jul 24 10:26 .
drwxr-xr-x. 1 jenkins jenkins 72 Jul 24 10:27 ..
drwxr-xr-x. 1 jenkins jenkins 136 Jul 24 10:26 .git
-rw-r--r--. 1 jenkins jenkins 100 Jul 24 10:26 README.md
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Test)
[Pipeline] sh
+ echo Running Tests...
Running Tests...
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] sh
+ echo Deployment Successful!
Deployment Successful!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

Jenkins 2.568.1

 Jenkins

+ New Item

Build History

Build Queue

No builds in the queue.

Build Executor Status

0/2

All

+

S	W	Name ↓	Last Success	Last Failure	Last Duration	
✓	☁	Git-Practice-Build	3 min 54 sec #3	5 min 32 sec #2	0.85 sec	▶
✓	☁	My-CICD-Pipeline	17 min #3	21 min #1	2.7 sec	▶

Icon: S M L

🔍 ⚙️ 👤

Add description